STE General Workflow — User Guide

This guide walks you through configuring and running the STE General Workflow, which generates a comprehensive elephant movement analysis report for Save the Elephants sourced from EarthRanger.

Overview

The workflow processes GPS observations for a chosen subject group and produces a full suite of spatial, temporal, and ecological analyses:

Movement maps - A **Movement Tracks Map** — current and previous period trajectories overlaid together, coloured by period (current vs. previous) for direct comparison - A **Subject Tracks Map** — trajectories coloured per individual subject using EarthRanger hex colours

Home range maps - An **Overall Home Range Map** — Elliptical Time Density (ETD) at percentiles 50–99.9%, coloured by the RdYlGn palette - A **Filtered Home Range Map** — 99th percentile home range only, shown in a single chocolate tone - A **Protected Areas Home Range Map** — ETD computed from fixes inside protected land (Community Conservancies, National Reserves, National Parks) - An **Unprotected Areas Home Range Map** — ETD computed from fixes outside protected land - **Seasonal Home Range Maps** (per subject) — 99.9th percentile ETD split into wet and dry seasons, coloured by season

Habitat use maps - A **Recursion Events Raster** — ecograph network weight raster classified into 6 natural-break bins, green → red - A **Dry Season Mean Speed Raster** — mean movement speed raster for dry-season fixes, 6 natural-break bins, green → red - A **Wet Season Mean Speed Raster** — same as above for wet-season fixes

Activity maps - A **Time of Day Dominance Map** — grid showing whether each cell is used predominantly at night (dark blue) or during the day (cornflower blue) - A **Night Fixes Map** — grid cells where ≥ 65% of fixes are nocturnal

Event maps - An **Elephant Sightings / Collaring Events Map** — scatter plot of EarthRanger events for the analysis period

Charts and tables - A **Protection Status Bar Chart** — proportion of fixes inside vs. outside protected areas - A **Seasonal Windows CSV** — NDVI-derived wet/dry season date ranges used to label trajectories

Reports - A **Word document report** (`.docx`) — cover page (logo, subject count, report period) plus a general per-subject section - An **interactive dashboard**

Prerequisites

Before running the workflow, ensure you have:

- Access to an **EarthRanger** instance containing the subject group to analyse
- Access to a **Google Earth Engine** service account (for NDVI-based seasonal window detection and satellite base layers)
- The **landDx geodatabase** (`.gpkg`) — the standard STE Dropbox URL is pre-filled; supply a custom URL or local path if needed

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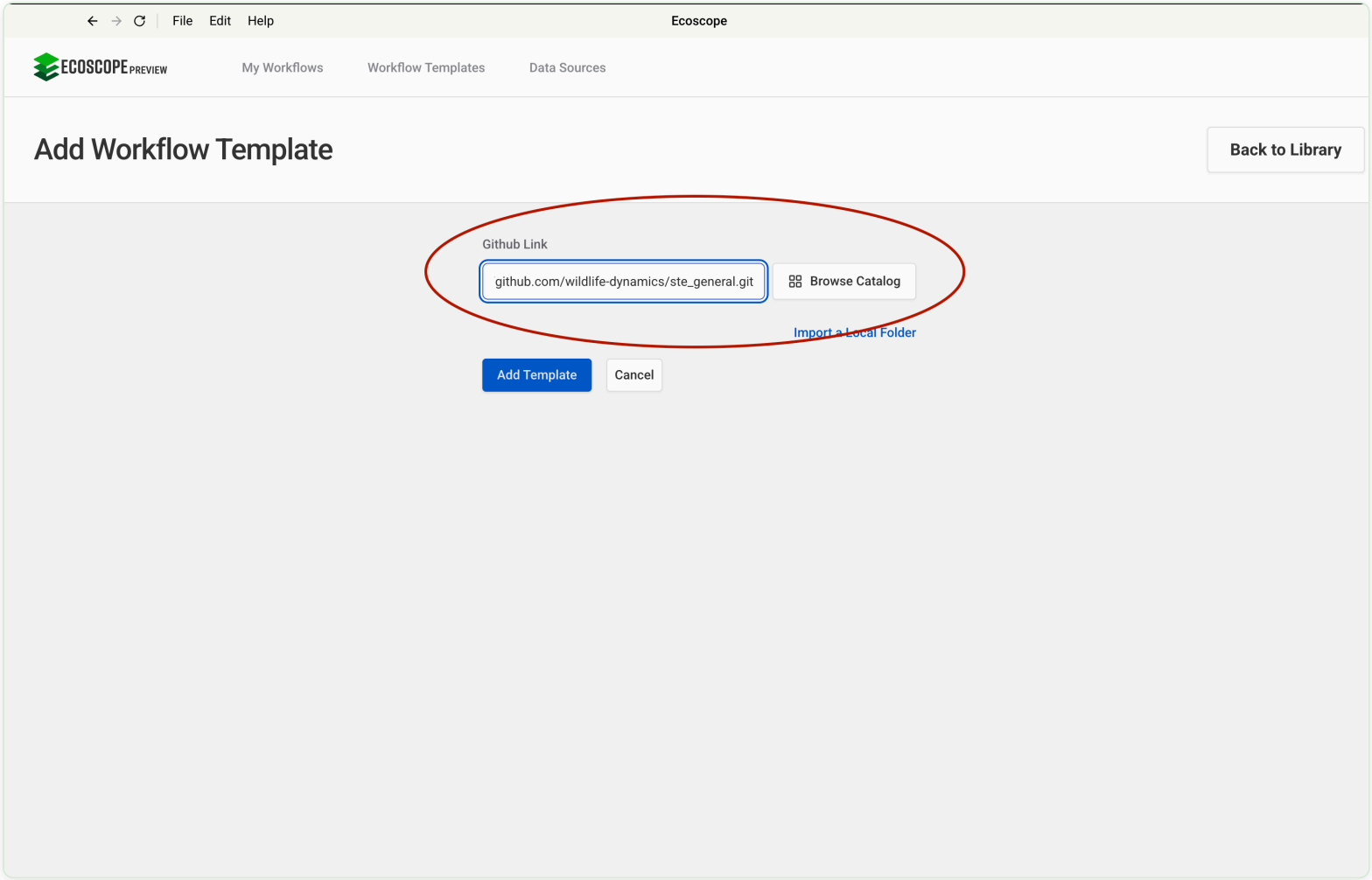
Step-by-Step Configuration

Step 1 — Add the Workflow Template

In the workflow runner, go to **Workflow Templates** and click **Add Workflow Template**. Paste the GitHub repository URL into the **Github Link** field:

```
https://github.com/wildlife-dynamics/ste_general.git
```

Then click **Add Template**.



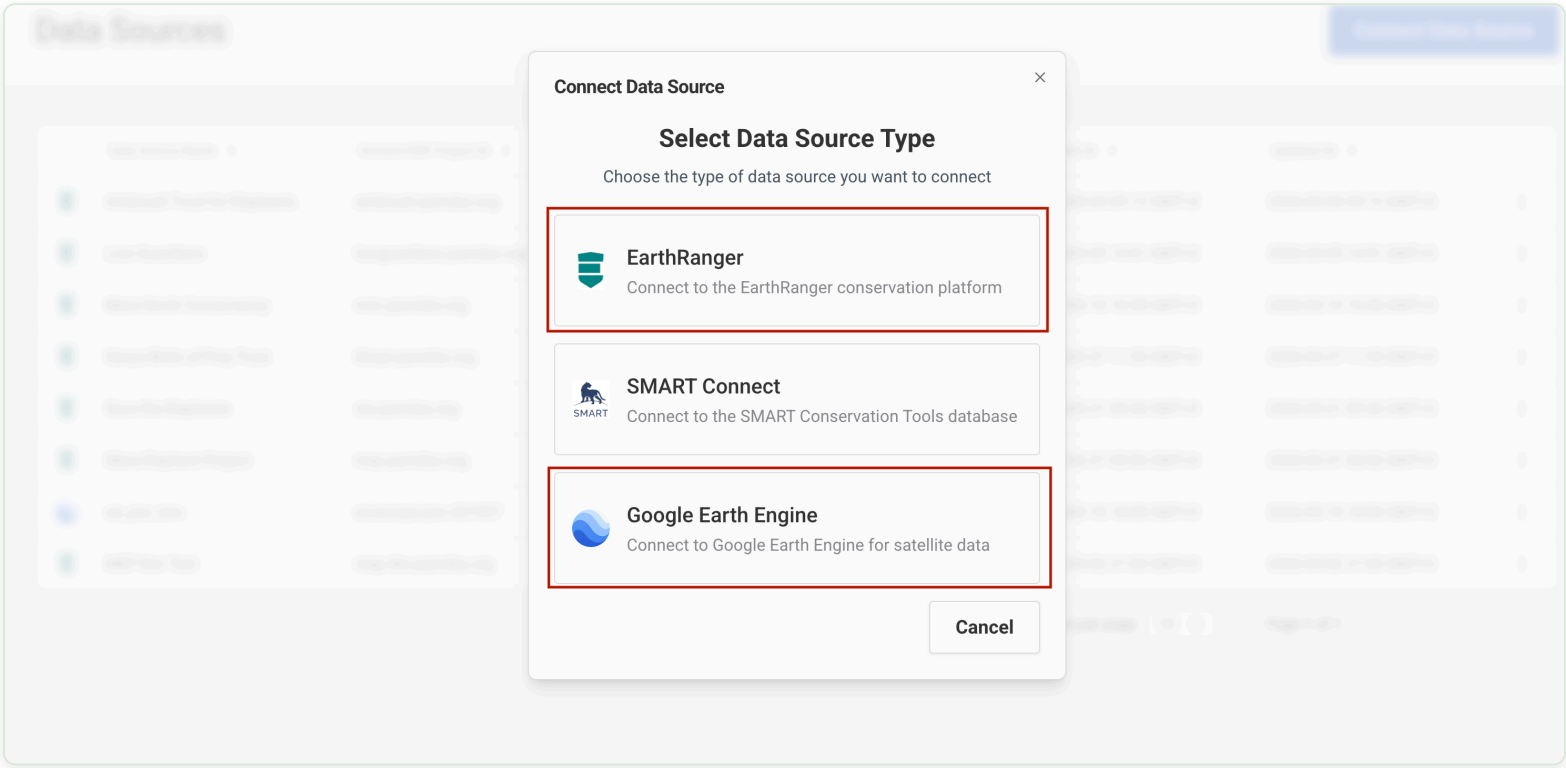
Add Workflow Template

Step 2 — Configure Connection

Navigate to **Data Sources** and click **Connect**. A dialog will appear prompting you to **Select Data Source Type**. This workflow requires two connections:

Data Source Type	Purpose
EarthRanger	Pull subject observations and collaring events
Google Earth Engine	Fetch satellite imagery and compute NDVI-based seasonal windows

Select **EarthRanger** first and complete Step 3, then repeat and select **Google Earth Engine** for Step 4.



Configure Connection

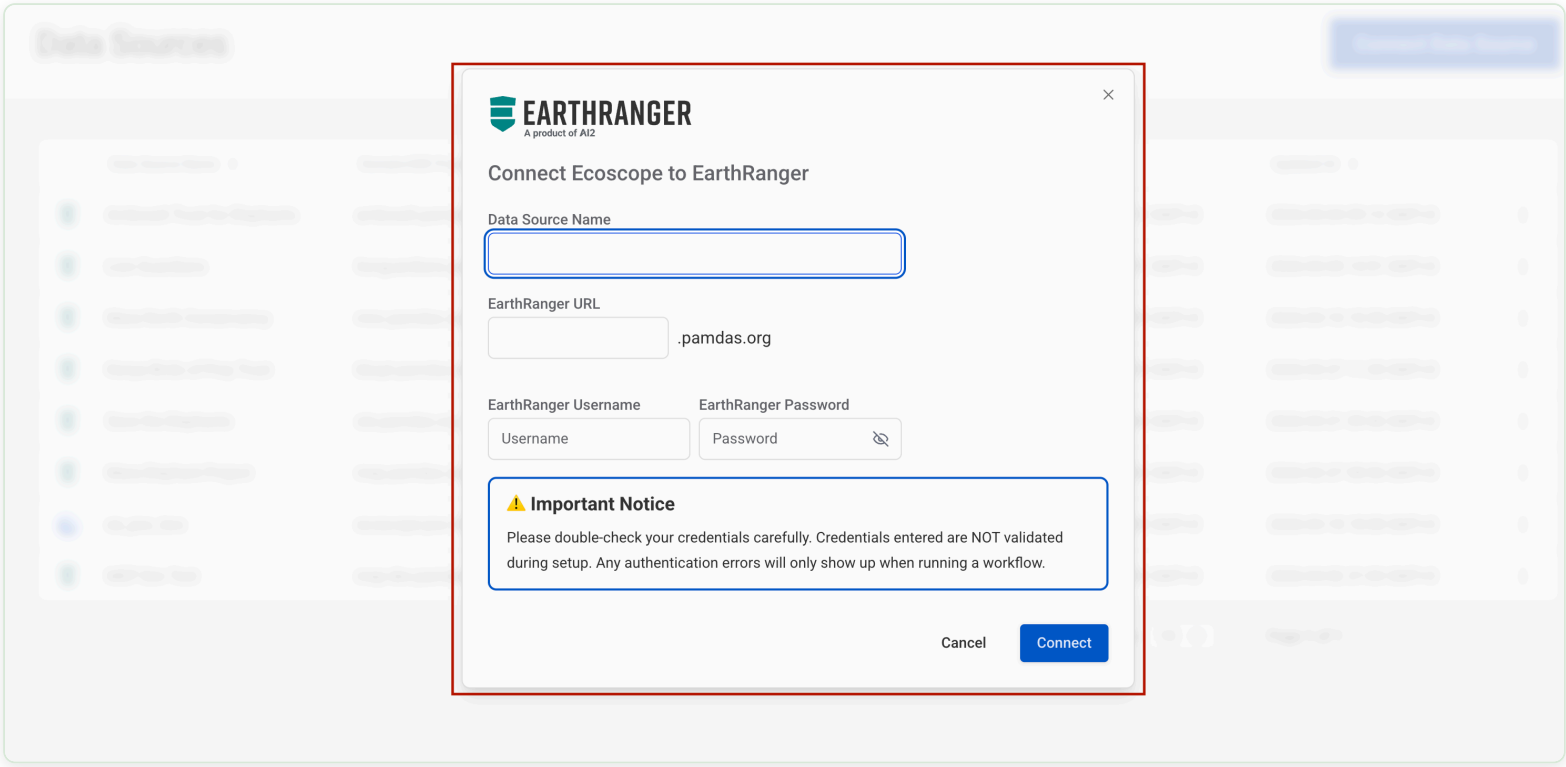
Step 3 — Add an EarthRanger Connection

After selecting **EarthRanger** in Step 2, fill in the connection form:

- **Data Source Name** — a label to identify this connection
- **EarthRanger URL** — your instance URL (e.g. `your-site.pamdas.org`)
- **EarthRanger Username** and **EarthRanger Password**

> Credentials are not validated at setup time. Any authentication errors will appear when the workflow runs.

Click **Connect** to save.



EarthRanger Connection

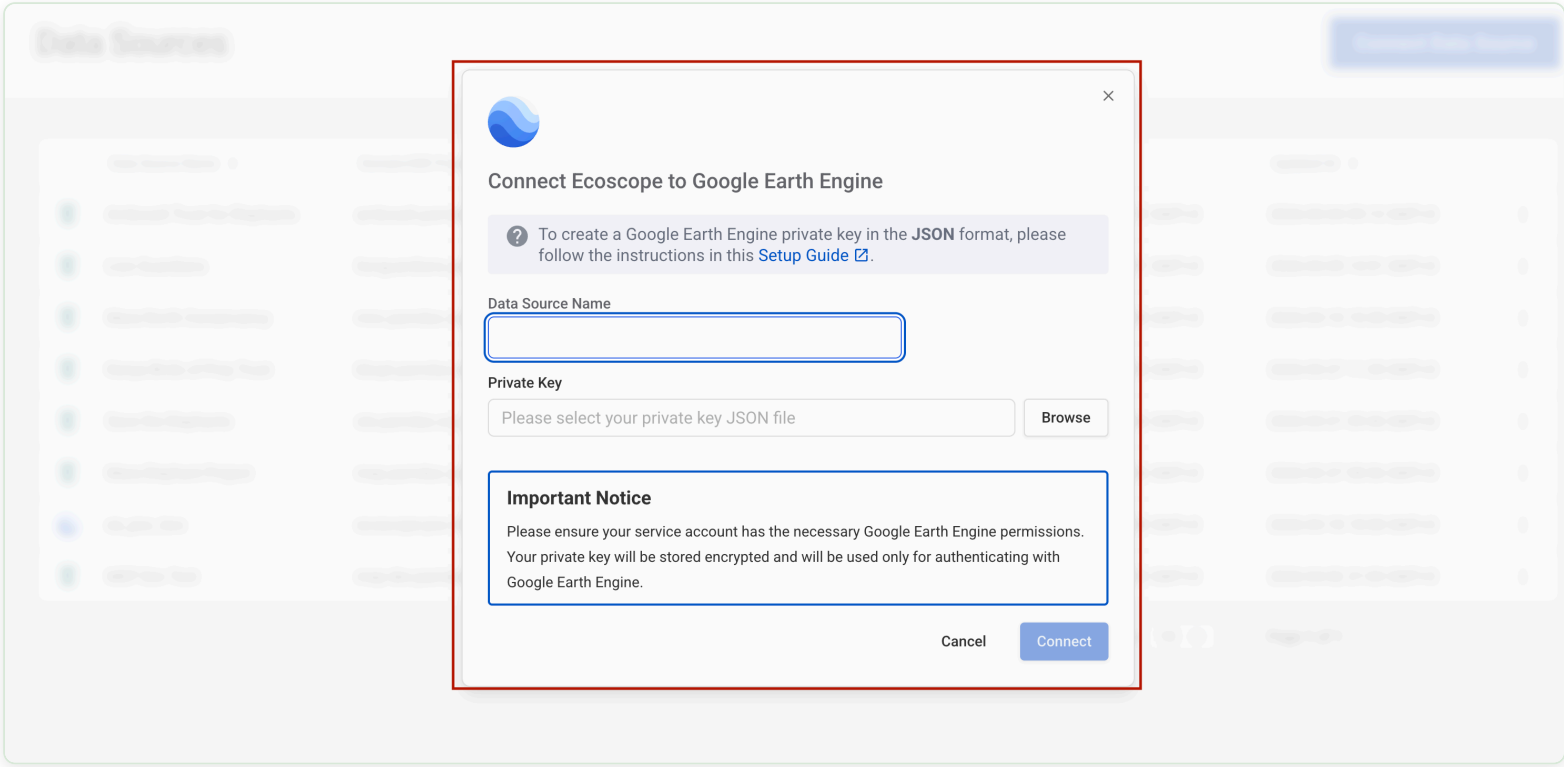
Step 4 — Add a Google Earth Engine Connection

After selecting **Google Earth Engine** in Step 2, fill in the connection form:

- **Data Source Name** — a label to identify this connection
- **Private Key** — click **Browse** to select your GEE service account private key file (JSON format)

> To generate a private key, follow the instructions in the Setup Guide. The key is stored encrypted and used only to authenticate with Google Earth Engine.

Click **Connect** to save.

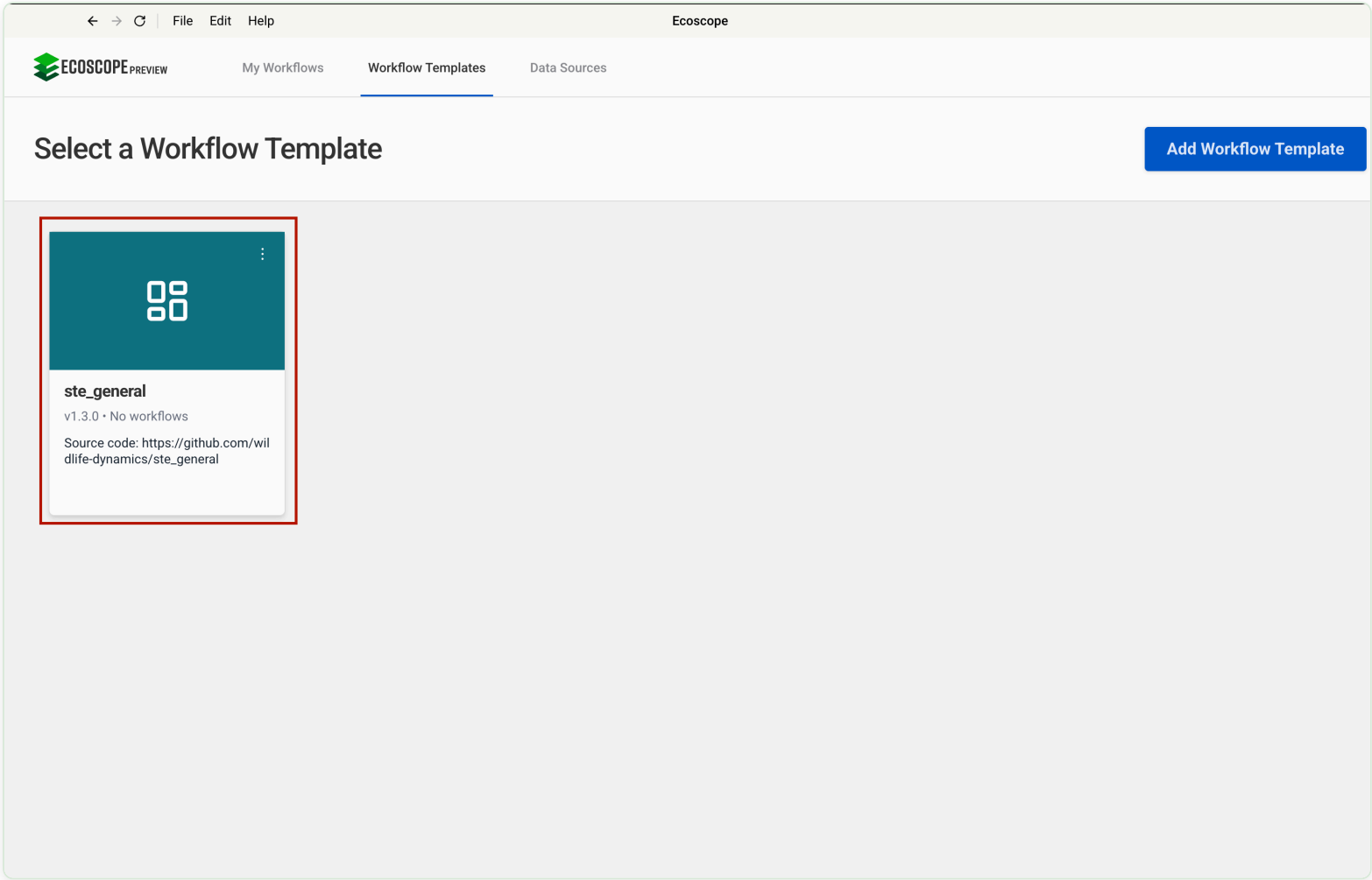


Google Earth Engine Connection

Step 5 — Select the Workflow

After the template is added, it appears in the **Workflow Templates** list as **ste_general**. Click it to open the workflow configuration form.

> The card may show **Initializing...** briefly while the environment is set up.



Select Workflow Template

Step 6 — Set Workflow Details and Define Analysis Time Range

The configuration form opens with two sections at the top.

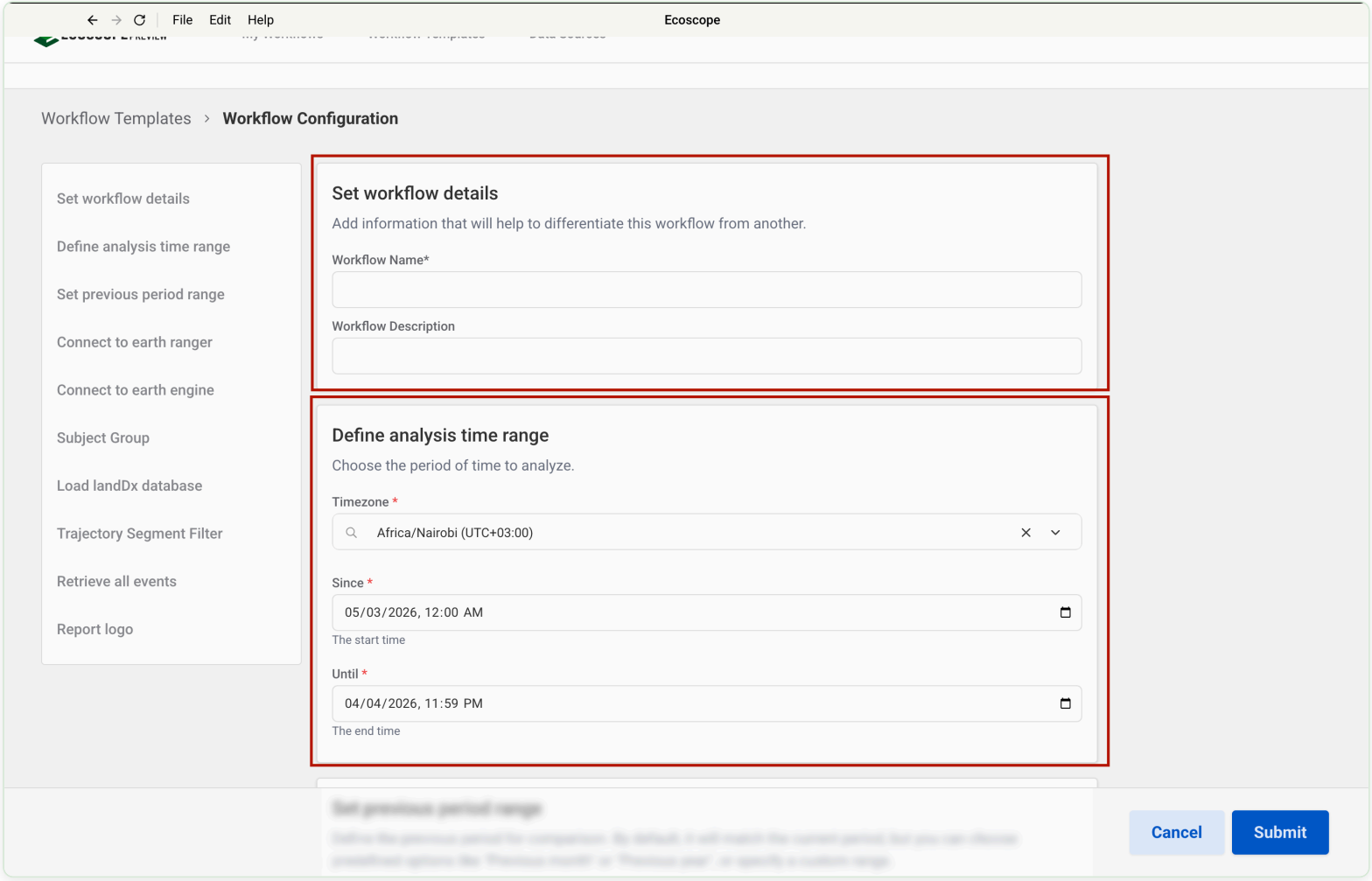
Set workflow details

Field	Description
Workflow Name	A short name to identify this run
Workflow Description	Optional notes (e.g. date range or subject group)

Define analysis time range

Field	Description
Timezone	Select the local timezone (e.g. `Africa/Nairobi UTC+03:00`)
Since	Start date and time of the analysis period
Until	End date and time of the analysis period

All subject tracks, home ranges, and activity maps are computed within this window.



Set Workflow Details and Define Analysis Time Range

Step 7 — Set Previous Period Range, Connect to EarthRanger, and Connect to EarthEngine

Scroll down to configure the next three sections.

Set previous period range

This period drives the “Previous tracks” layer on the Movement Tracks Map, enabling direct before-and-after comparison alongside the current period tracks.

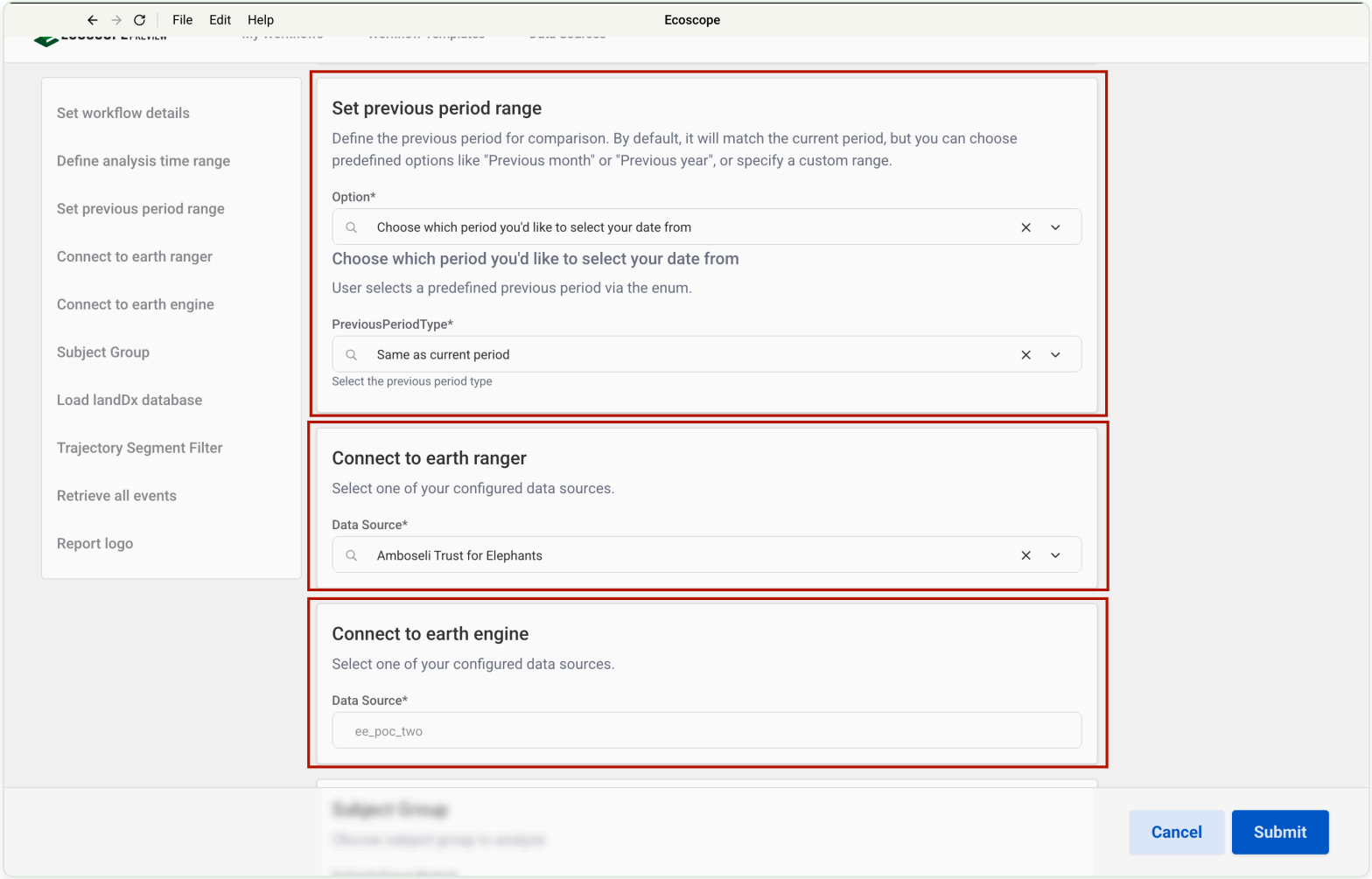
Field	Default	Description
Option	Choose which period to select	Whether to use a predefined period type or specify a custom date range
PreviousPeriodType	`Same as current period`	Predefined period (e.g. Same as current, Previous month, Previous year)

Connect to earth ranger

Select the EarthRanger data source configured in Step 3 from the **Data Source** dropdown.

Connect to earth engine

Select the Google Earth Engine data source configured in Step 4 from the **Data Source** dropdown. This connection is used to retrieve NDVI satellite data for detecting wet and dry seasonal windows, which are then used to label trajectories and generate seasonal home range and speed raster maps.



Set Previous Period Range, Connect to EarthRanger, and Connect to EarthEngine

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Step 8 — Subject Group and Load landDx Database

Subject Group

Field	Default	Description
Subject Group Name	`Elephants`	Name of the subject group to pull from EarthRanger

Load landDx database

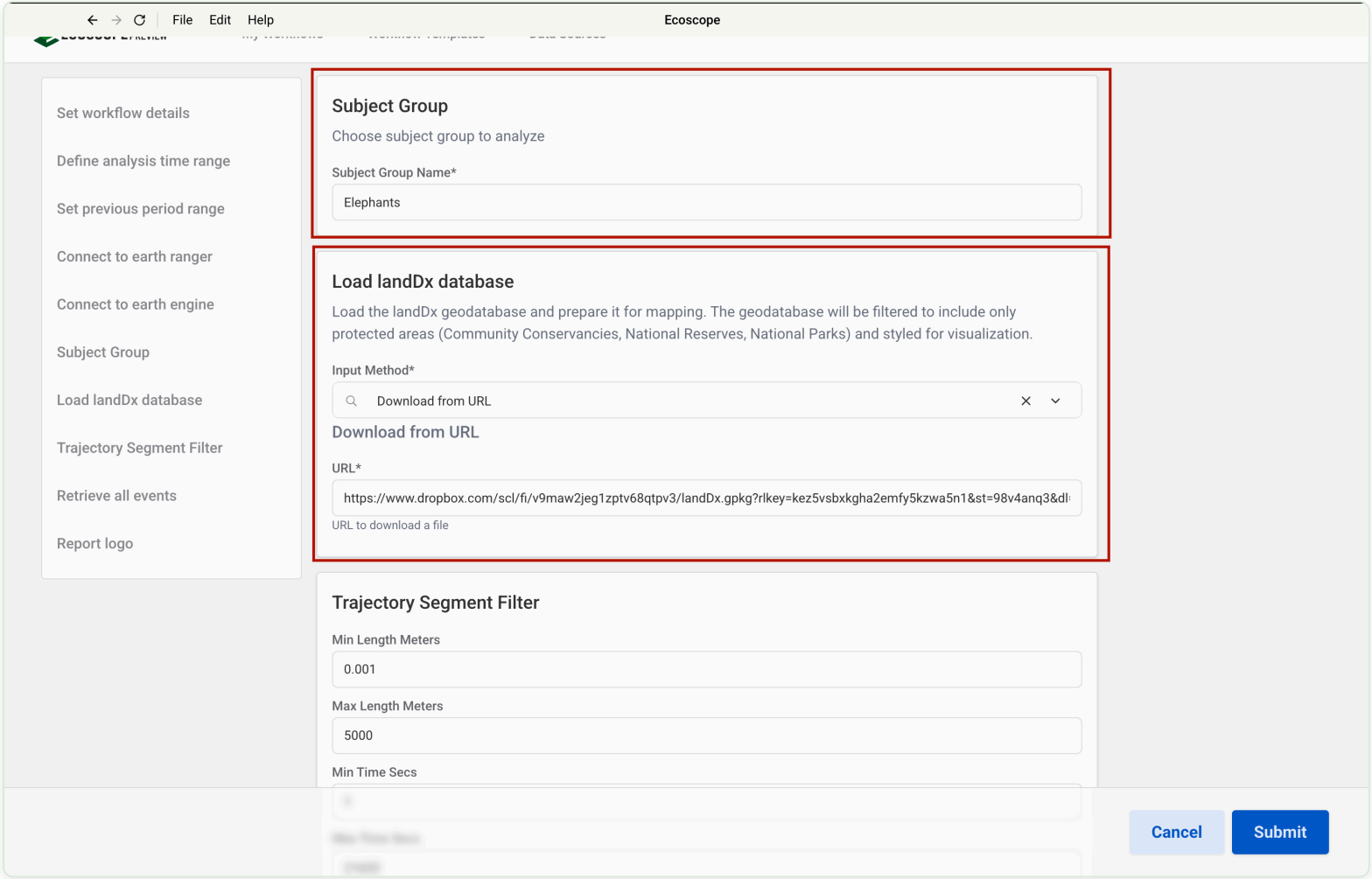
The landDx geodatabase supplies the protected area and conservancy boundary layers displayed on all maps. The workflow filters the database to three area types and styles them as follows:

Area Type	Map colour
Community Conservancy	`#a6b697` (sage green)
National Reserve	`#88a78e` (muted green)
National Park	`#115631` (dark forest green)

Choose how to supply the file:

Input Method	When to use	What to provide
Download from URL	File is hosted remotely (e.g. Dropbox)	Paste the direct download URL into the **URL** field
Local path	File is already on the machine running the workflow	Enter the absolute file path to the `.gpkg` file

> The standard STE landDx Dropbox URL is pre-filled. Leave it unchanged unless you have a custom or updated version of the file.



Subject Group and Load landDx Database

Step 9 — Trajectory Segment Filter, Retrieve All Events, and Report Logo

Trajectory Segment Filter

Applied to both the current and previous period trajectories before any analysis. Removes GPS noise and biologically unrealistic movement segments.

Field	Default	Description
Min Length Meters	`0.001`	Discard segments shorter than this distance
Max Length Meters	`5000`	Discard segments longer than this distance
Min Time Secs	`1`	Discard segments shorter than this duration
Max Time Secs	`21600`	Discard segments longer than this duration (6 hours)
Min Speed Knots	`0.01`	Discard segments below this average speed
Max Speed Knots	`9`	Discard segments above this average speed

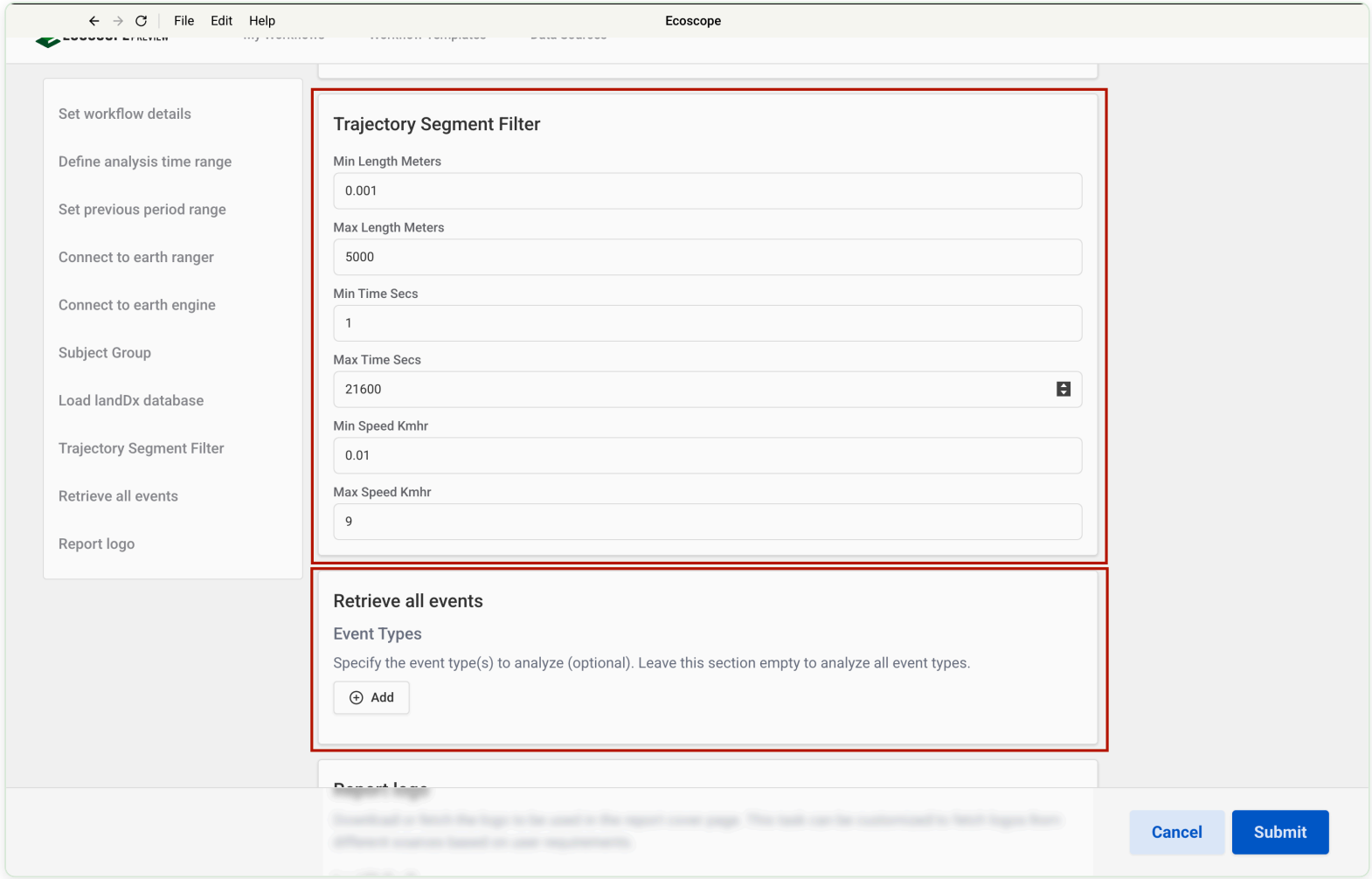
Retrieve all events

Field	Description
Event Types	Filter to specific EarthRanger event types (click **Add** to add entries; leave empty to include all event types)

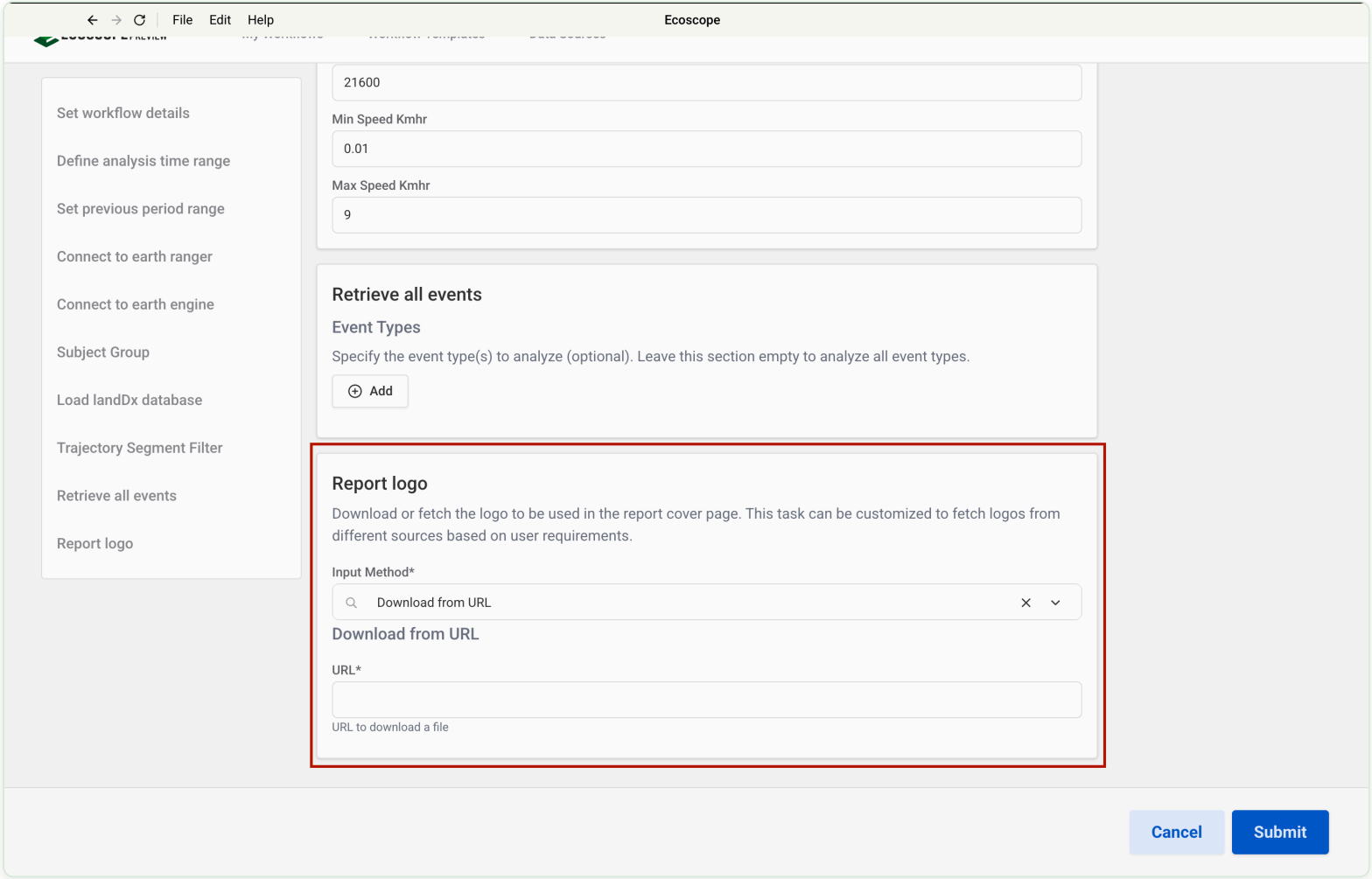
Report logo

Provide the logo image to appear on the report cover page.

Input Method	What to provide
Download from URL	Paste the direct download URL into the **URL** field



Trajectory Segment Filter and Retrieve All Events



Report Logo

Running the Workflow

Once all parameters are configured, click **Submit**. The runner will:

1. Pull subject GPS observations (current and previous period) from EarthRanger.

2. Process relocations and apply the trajectory segment filter to both periods.
3. Query Google Earth Engine to determine NDVI-derived wet and dry seasonal windows; label each trajectory segment accordingly.
4. Load and style the landDx geodatabase boundary layers.
5. Generate the Movement Tracks Map (current + previous overlay) and the Subject Tracks Map.
6. Fetch EarthRanger events and generate the Elephant Sightings map.
7. Compute Elliptical Time Density home ranges: overall, filtered 99th percentile, protected/unprotected split, and per-subject seasonal.
8. Generate the Recursion Events raster and dry/wet season Mean Speed rasters.
9. Compute the Day/Night Dominance grid and Night Fixes grid.
10. Render the Protection Status bar chart.
11. Assemble the Word report (cover page + general section) and the dashboard.
12. Save all outputs to the directory specified by `ECOSCOPE_WORKFLOWS_RESULTS`.

Output Files

All outputs are written to `$ECOSCOPE_WORKFLOWS_RESULTS/`:

File	Description
`relocations.geoparquet`	Current period GPS fix locations with day/night labels
`previous_period_relocations.geoparquet`	Previous period GPS fix locations
`trajectories.geoparquet`	Current period trajectory segments with season labels
`previous_period_trajectories.geoparquet`	Previous period trajectory segments
`home_range_etd.geoparquet`	Overall ETD home range polygons (50–99.9th percentiles)

File	Description
`protected_areas_home_range_etd.geoparquet`	ETD home range for fixes inside protected areas
`unprotected_areas_home_range_etd.geoparquet`	ETD home range for fixes outside protected areas
`<subject>_seasonal_etd.geoparquet`	Seasonal ETD per subject
`seasonal_windows.csv`	NDVI-derived wet/dry season date ranges
`weighted_raster.*`	Recursion events ecograph raster file
`movement_tracks.html`	Interactive movement tracks map (current + previous period)
`subject_tracks.html`	Interactive subject tracks map (coloured per individual)
`collared_points.html`	Interactive elephant sightings / collaring events map
`overall_homerange.html`	Interactive overall home range map
`filtered_homerange.html`	Interactive 99th percentile home range map
`protected_areas.html`	Interactive protected areas home range map
`unprotected_areas.html`	Interactive unprotected areas home range map
`<subject>_seasonal_homerange.html`	Interactive seasonal home range map per subject
`recursion_events.html`	Interactive recursion events raster map
`dry_mean_speed_raster_map.html`	Interactive dry season mean speed raster map
`wet_mean_speed_raster_map.html`	Interactive wet season mean speed raster map
`time_of_day_dominance.html`	Interactive time of day dominance grid map

File	Description
`night_fixes.html`	Interactive night fixes grid map
`protection_status_bar.html`	Interactive protection status bar chart
`movement_tracks.png`	Screenshot of movement tracks map (2× resolution)
`subject_tracks.png`	Screenshot of subject tracks map (2× resolution)
`collared_points.png`	Screenshot of collaring events map (2× resolution)
`overall_homerange.png`	Screenshot of overall home range map (2× resolution)
`filtered_homerange.png`	Screenshot of filtered 99th percentile home range map (2× resolution)
`protected_areas.png`	Screenshot of protected areas home range map (2× resolution)
`unprotected_areas.png`	Screenshot of unprotected areas home range map (2× resolution)
`<subject>_seasonal_homerange.png`	Screenshot of seasonal home range per subject (2× resolution)
`recursion_events.png`	Screenshot of recursion events raster map (2× resolution)
`dry_mean_speed_raster_map.png`	Screenshot of dry season speed raster map (2× resolution)
`wet_mean_speed_raster_map.png`	Screenshot of wet season speed raster map (2× resolution)
`time_of_day_dominance.png`	Screenshot of time of day dominance map (2× resolution)
`night_fixes.png`	Screenshot of night fixes map (2× resolution)
`protection_status_bar.png`	Screenshot of protection status bar chart (2× resolution)
`general_cover_page.docx`	Rendered report cover page (logo, subject count, report period)

File	Description
`general.docx`	General per-subject report section
Merged report `.docx`	Final combined Word report